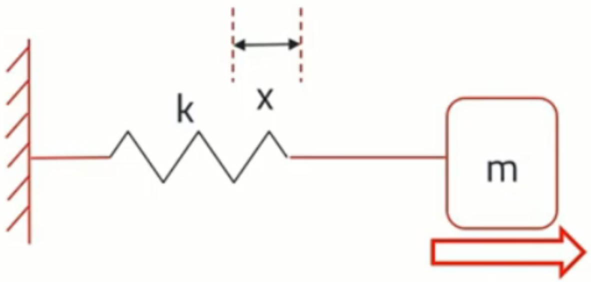
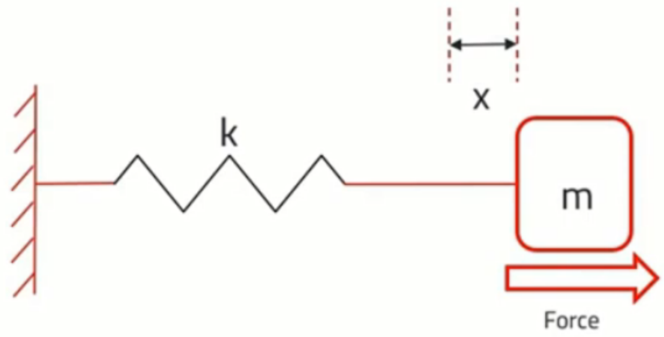
Acceleration

Accelerometer

GENERAL PRINCIPAL OF ACCELERATION





Hooke's law

F = kx

F - Force due to spring tension

k - Spring constant

x - Extension of the spring

m - Mass of the Object

F = kx = ma

a = k/m (x)

Newton's second law

F = ma

F - Force applied on the object

m - Mass of the object

a - Acceleration

x - Displacement of the object

Accelerometer

DISPLACEMENT MEASUREMENT

$$m\ddot{x} + c\dot{x} + kx = -m\ddot{a}_{ext}$$

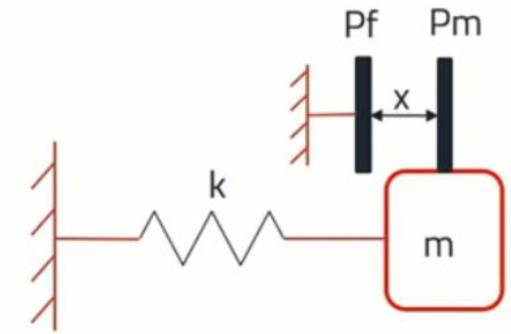
Different Technique:

Resistive

Inductive

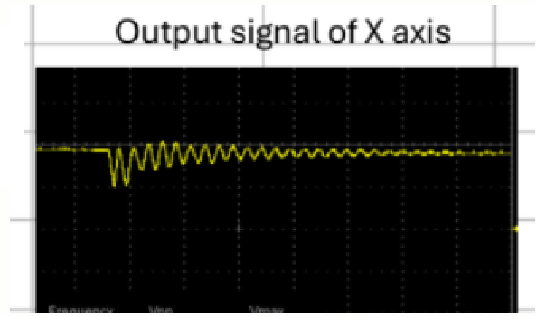
Capacitive

Capacitive Technique:



$C = \varepsilon_0 A / (x)$

$a = k / m\ (x)$



$$Q = \frac{f_n}{\Delta f}$$

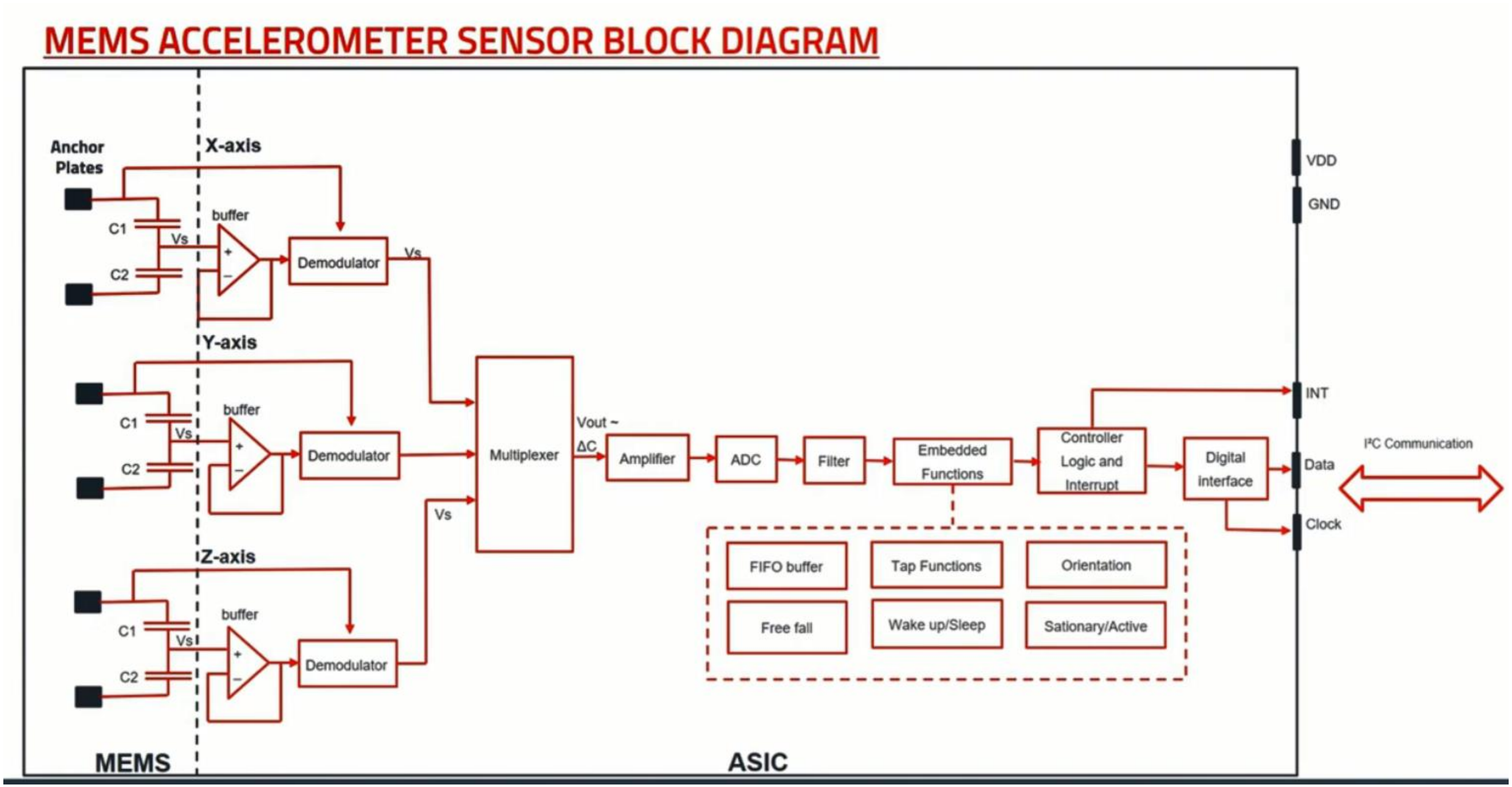
$$\zeta = \frac{1}{2Q}$$

Pm - Movable capacitor plate

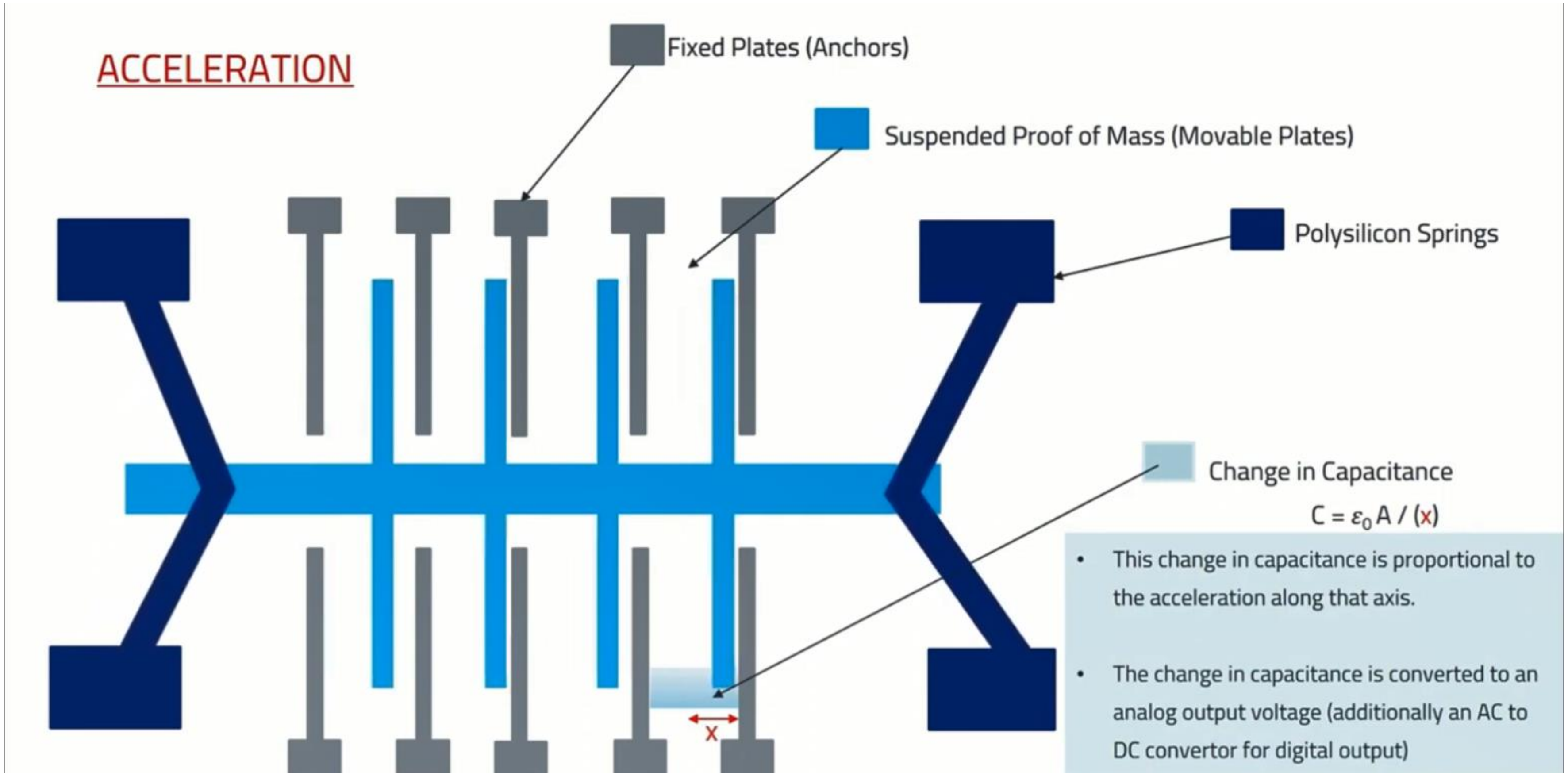
Pf - Fixed capacitor plate

x - Displacement

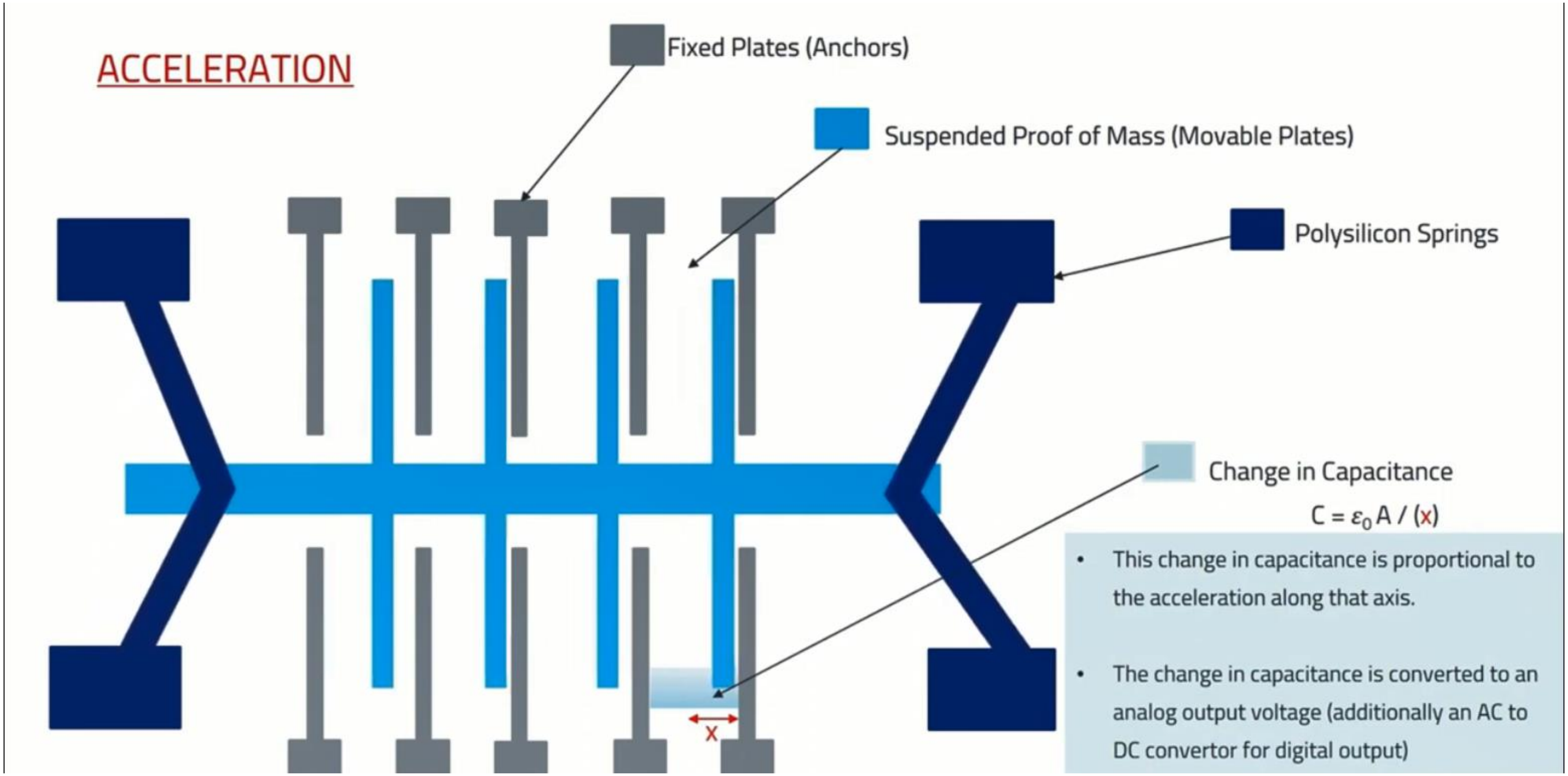
Accelerometer



Accelerometer



Accelerometer



Piezoelectric acceleration sensors

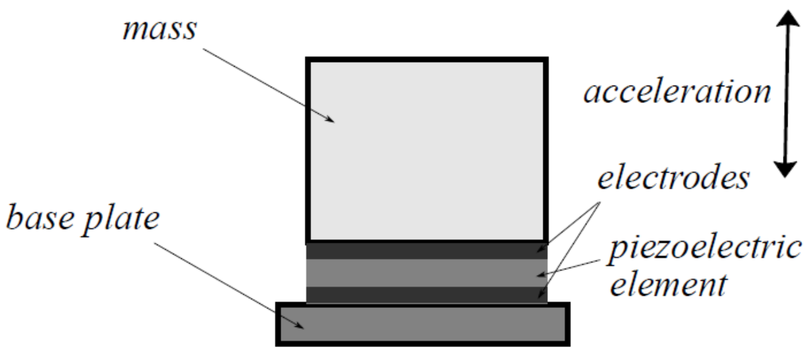


FIGURE 9.2: Principle of piezoelectric acceleration sensor

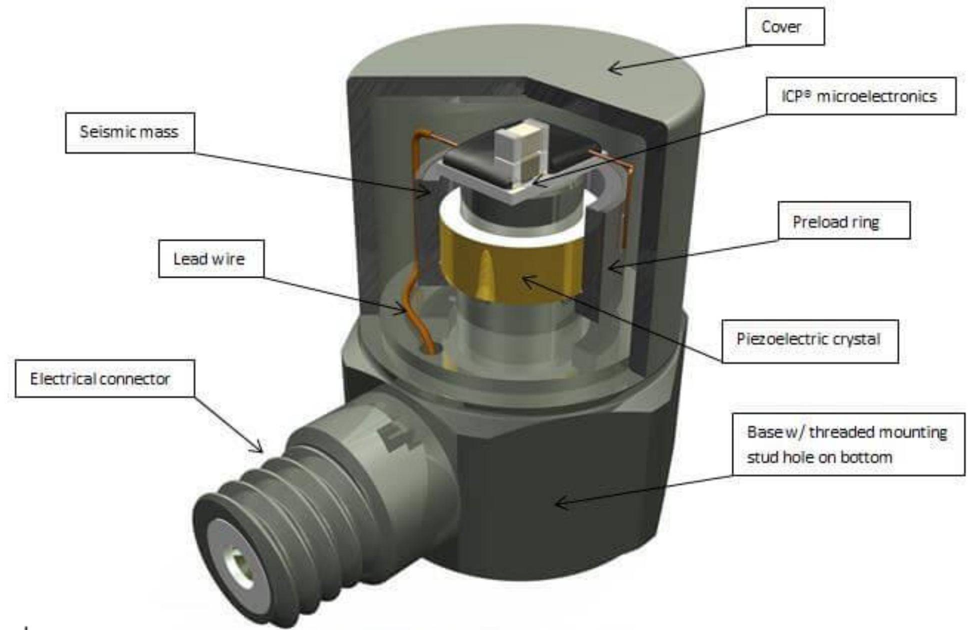
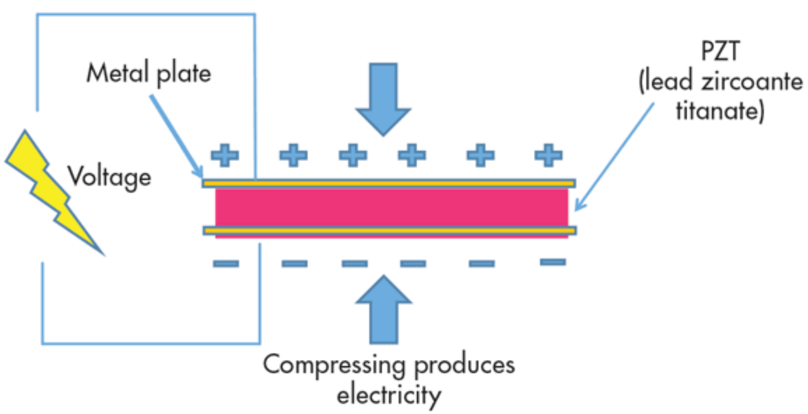


Figure 1: Typical ICP® Accelerometer



Piezoresistive acceleration sensors

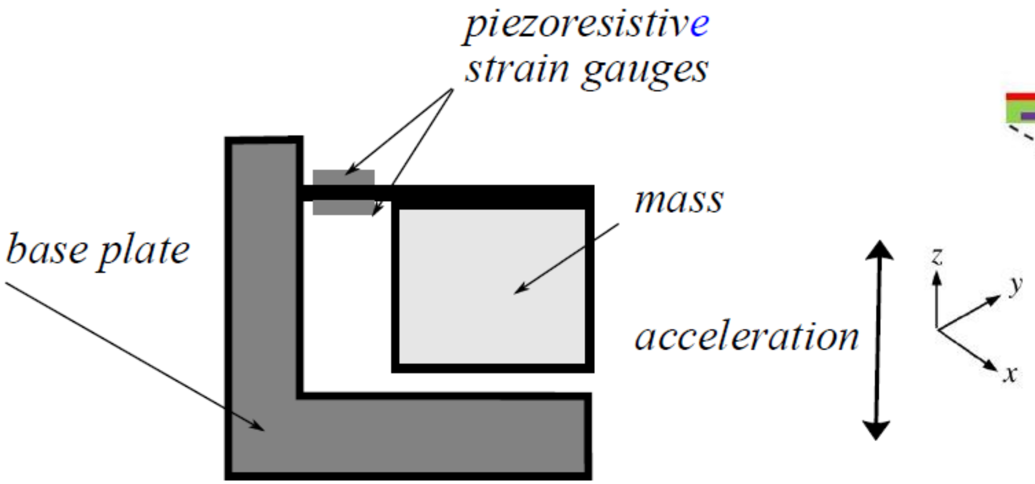
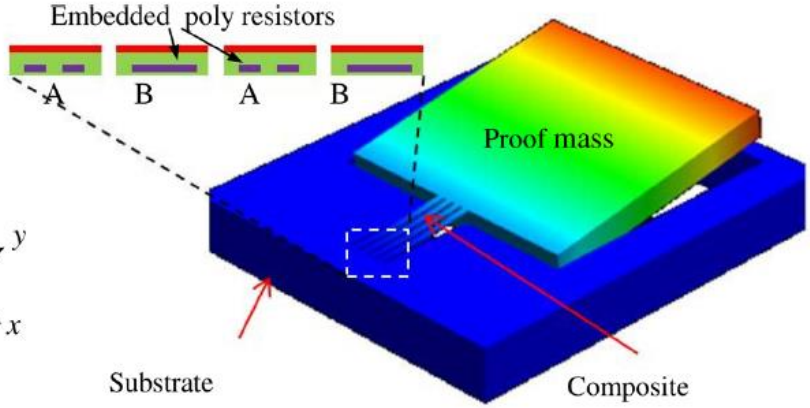
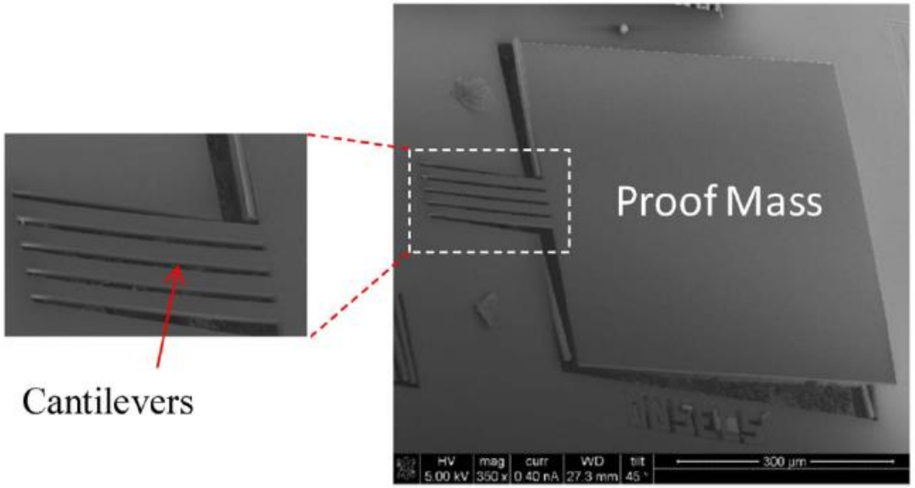


FIGURE 9.5: Piezoresistive acceleration sensor





Acceleration sensors with measured displacement

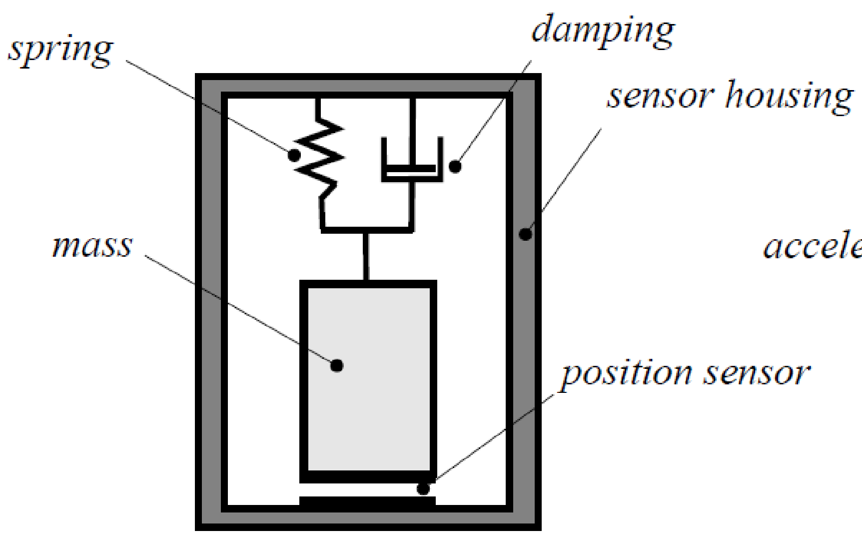


FIGURE 9.8: General principle of acceleration sensors with measured displacement

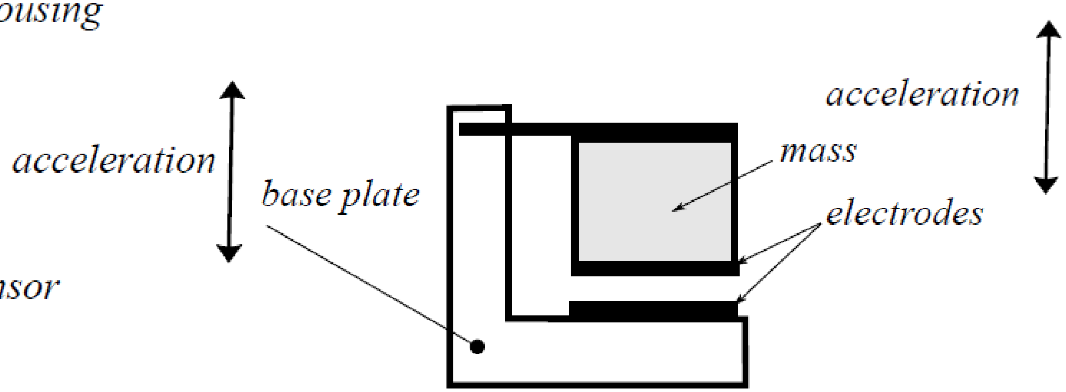
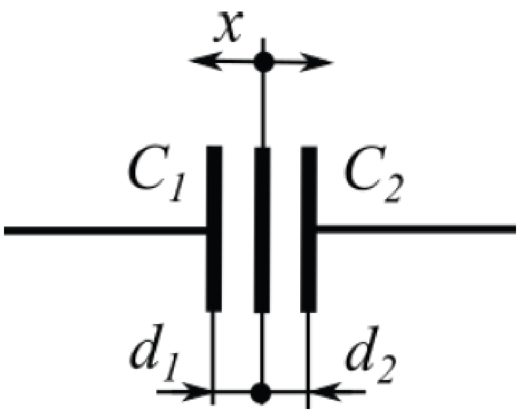
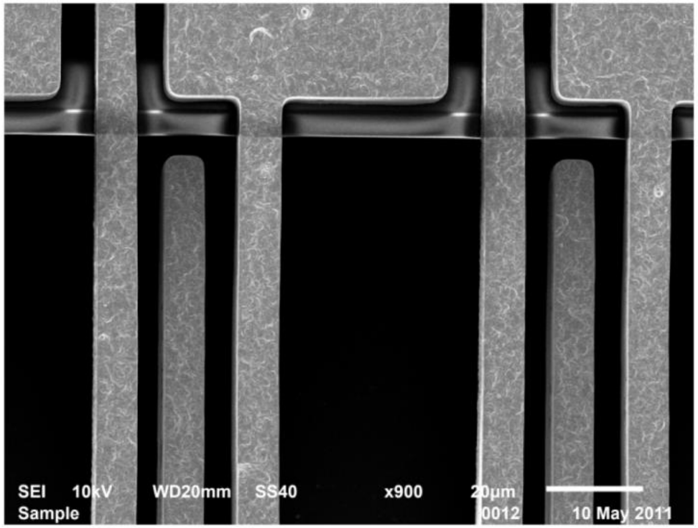
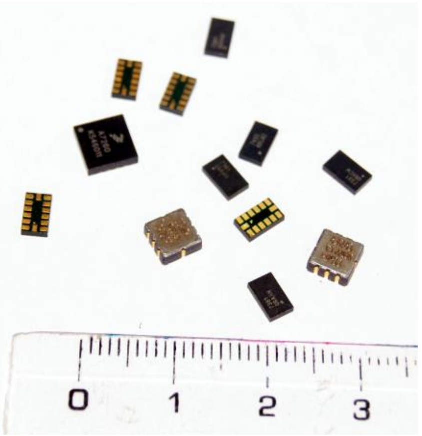


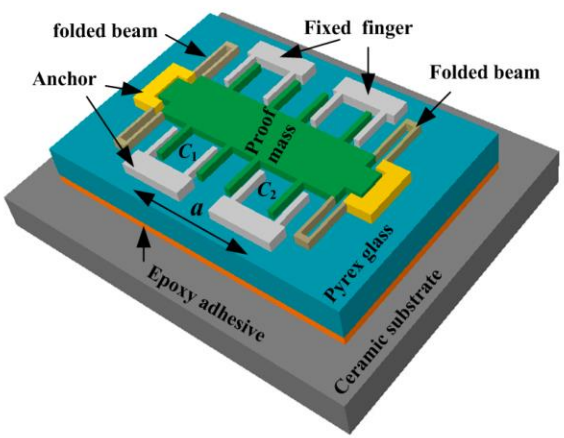
FIGURE 9.9: Capacitive accelerometer

Acceleration sensors with measured displacement

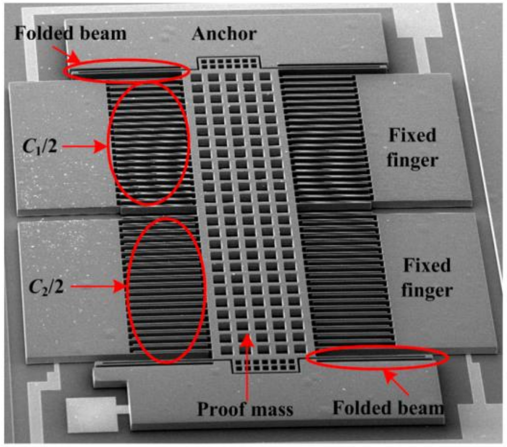






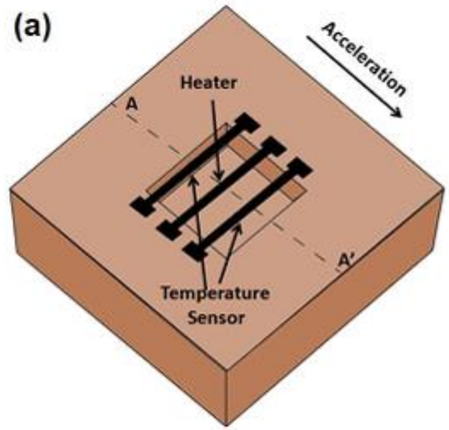


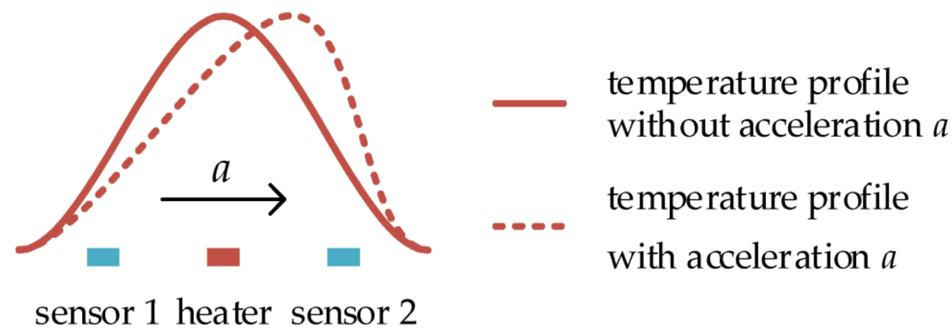
(a)

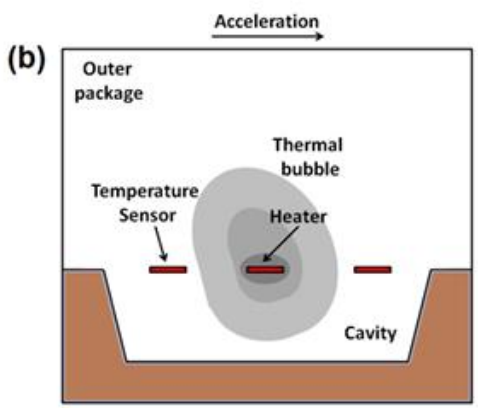


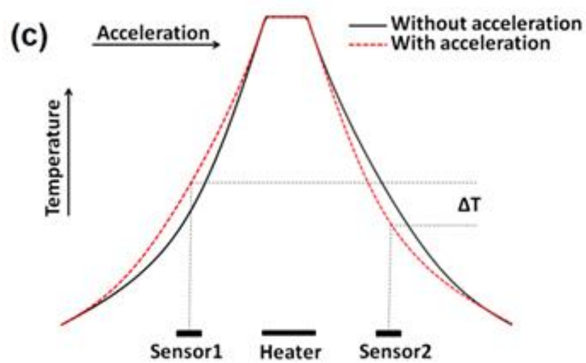
(b)

Thermal gas gyroscope





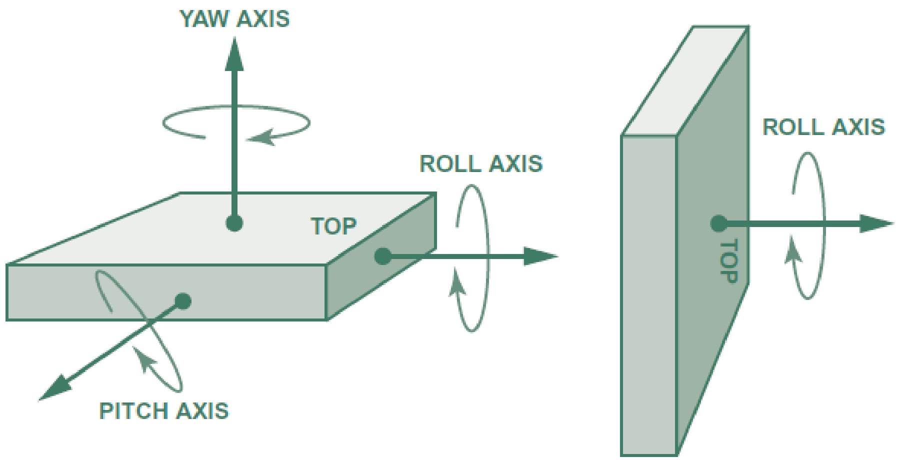




- Mukherjee, Rahul, et al. "A review of micromachined thermal accelerometers." *Journal of Micromechanics and Microengineering* 27.12 (2017): 123002.
- Liu S, Zhu R. Micromachined Fluid Inertial Sensors. *Sensors*. 2017; 17(2):367. https://doi.org/10.3390/s17020367

Gyroscope

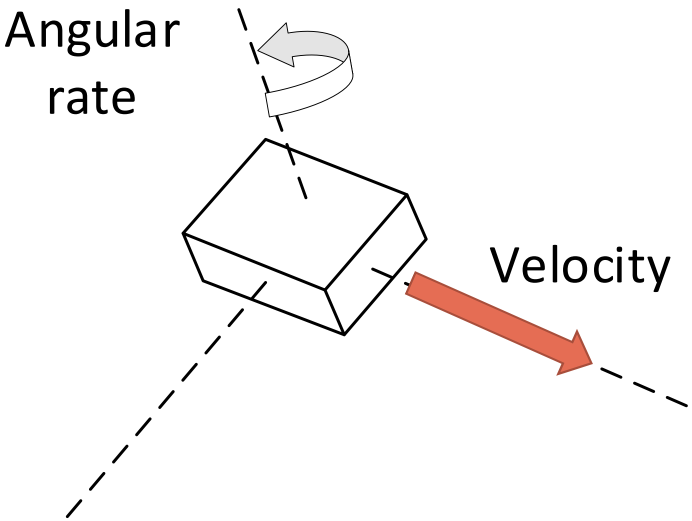
- Gyroscope is a type of inertial sensor that measures the angular rate

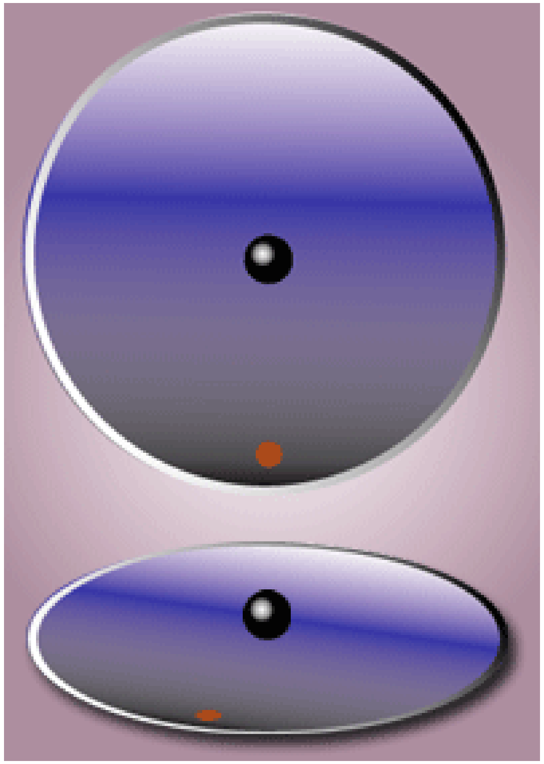


- Applications:
  - Measure how quickly an object turns
  - Integrate the angular rate over time to determine angular position

Gyroscope

- Conventional Vibratory Gyroscope:
  - Measure angular rate by means of Coriolis effects



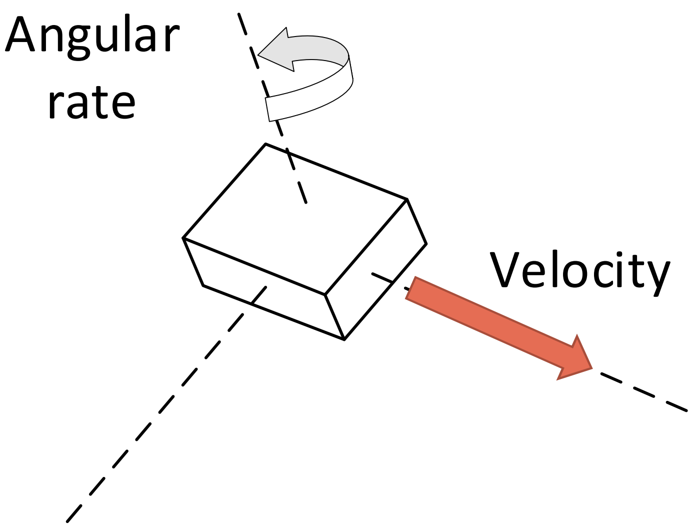


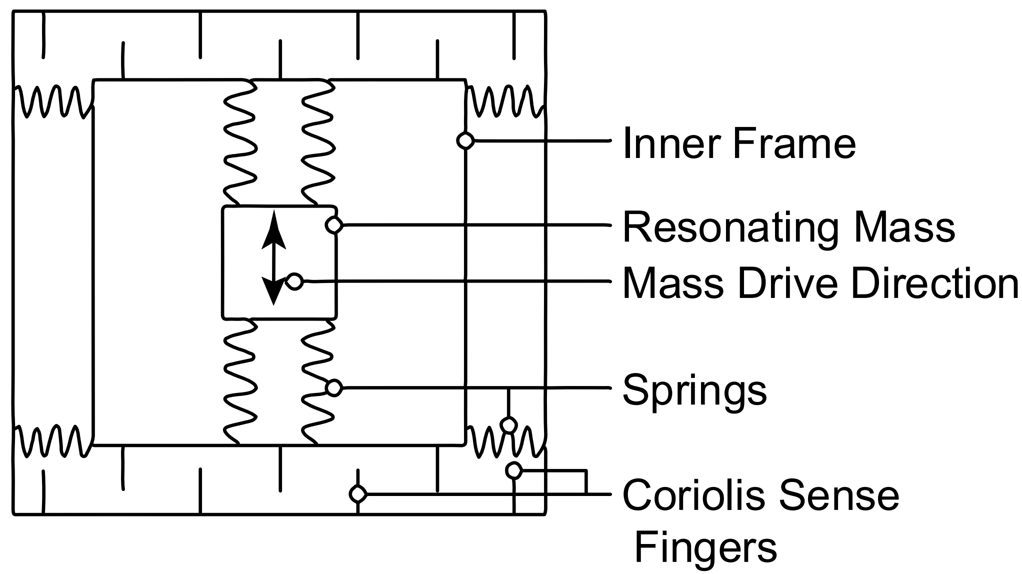
$F_C = -2m(\Omega \times v)$

- m: mass
- Ω: angular velocity
- v: tangential velocity

Gyroscope

- Conventional Vibratory Gyroscope:
  - Measure angular rate by means of Coriolis effects





$$F_C = -2m(\Omega \times v)$$

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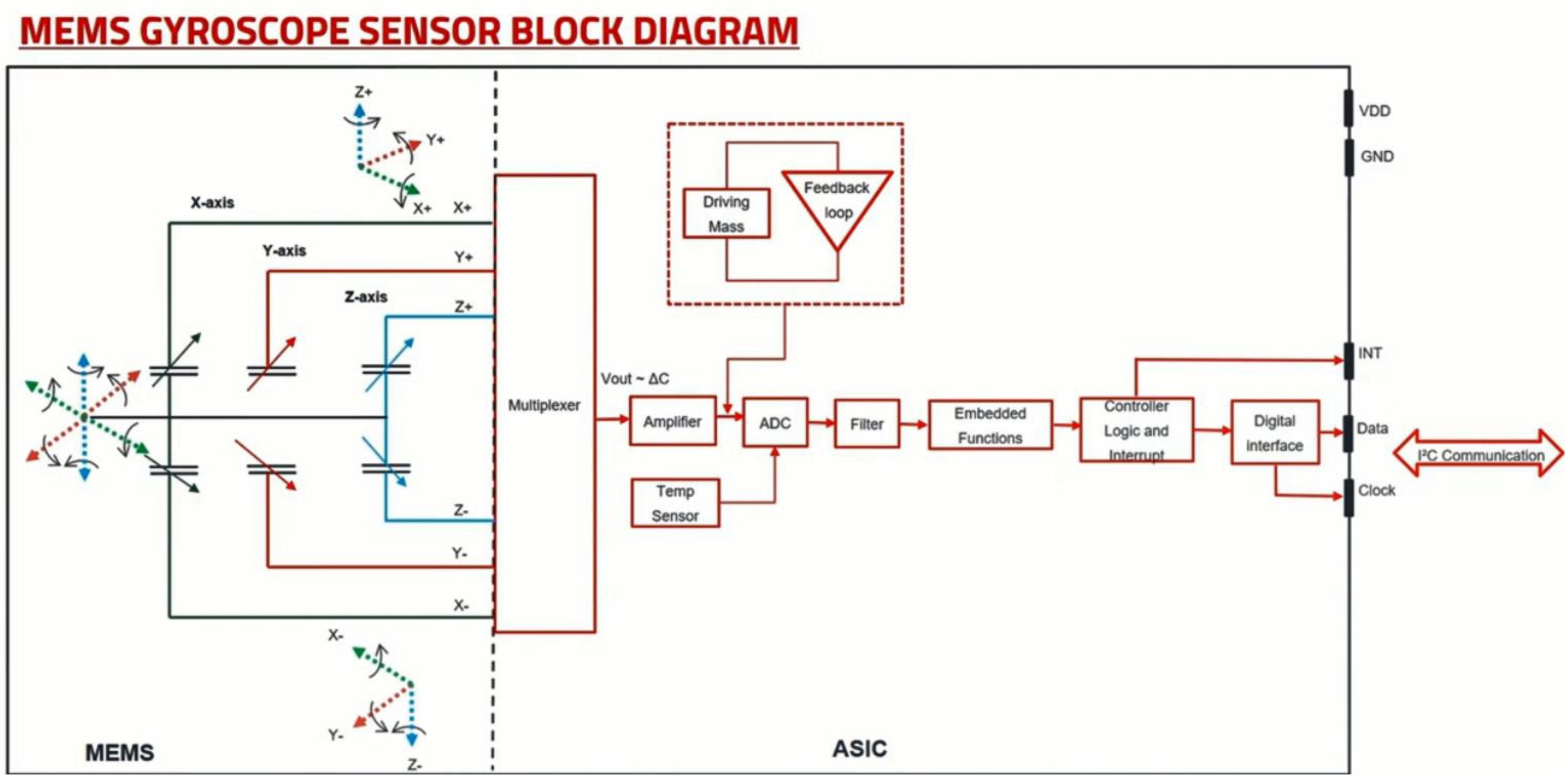
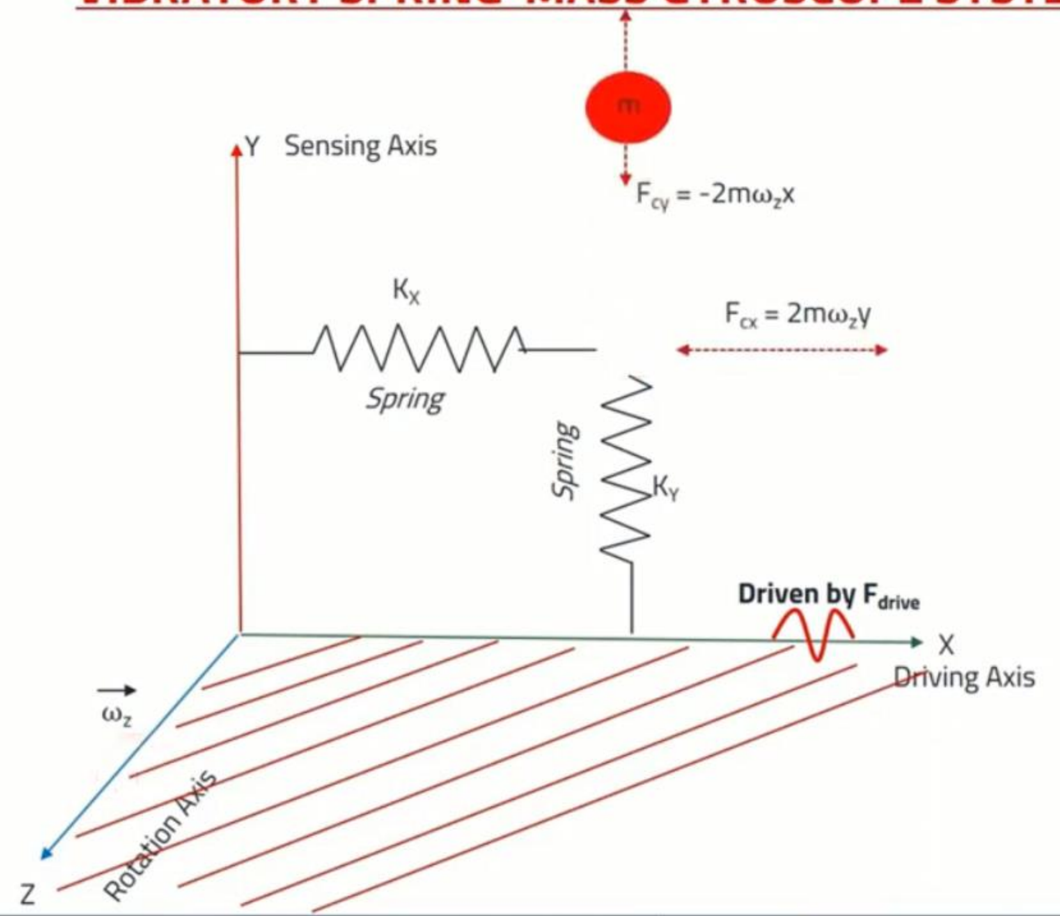
Gyroscope



MEMS GYROSCOPE SENSOR BLOCK DIAGRAM
Z+
Y+
X+
X+
X-axis
Y-axis
Y+
Z-axis
Z+
Driving Mass
Feedback loop
VDD
GND
INT
Vout ~ ΔC
Multiplexer
Amplifier
ADC
Filter
Embedded Functions
Controller Logic and Interrupt
Digital interface
Data
I²C Communication
Clock
Temp Sensor
Z-
Y-
X-
X-
Y-
Z-
MEMS
ASIC

Gyroscope

VIBRATORY SPRING-MASS GYROSCOPE SYSTEM



- m is the mass in X-Y plane
- K_x (K_y) - spring-constants
- X represents the driving axis
- Y represents the sensing axis
- Z represents the rotation axis
- F_{drive} → Ref driving signal (freq and amplitude)

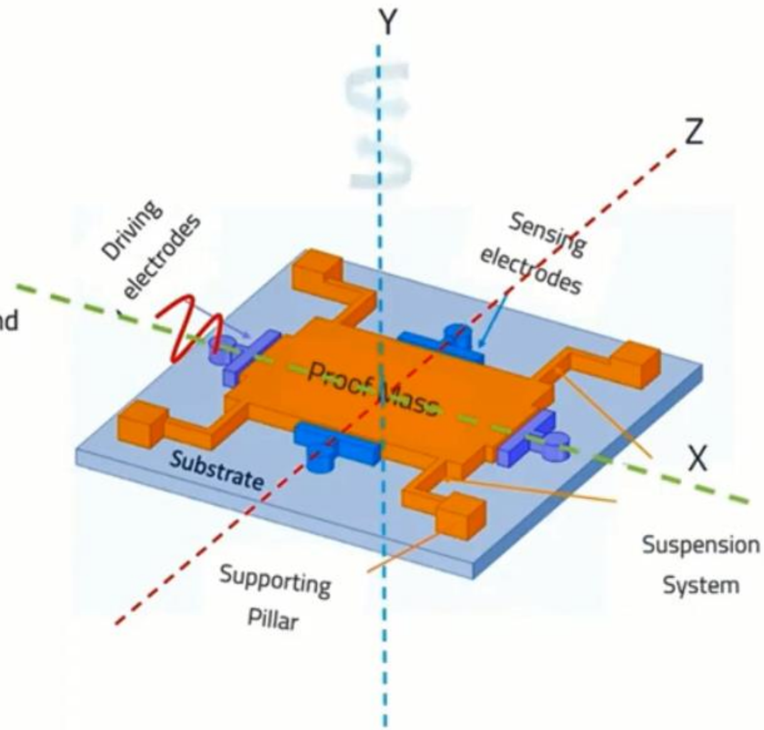
$\omega_z > 0$ (angular rate applied along Z axis)

- F_{cx} and F_{cy} →Coriolis force appear in X & Y axis

Gyroscope

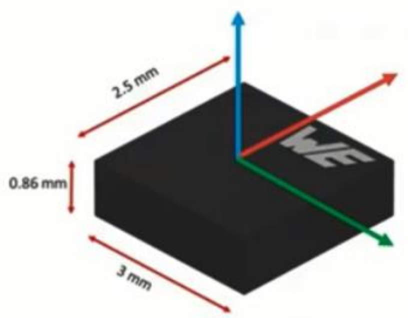
GYROSCOPE MEMS IMPLEMENTATION

- A Proof mass (m) placed above the silicon substrate.
- Driving, sensing electrodes, and a suspension support beam system with four fixed support pillars.
- The driving system provides an oscillation (X-axis) with a certain amplitude and frequency.
- The **proof mass becomes an oscillator**
- The same proof mass upon external rotation (Y-axis) changes its oscillation movement to the orthogonal direction (Z-axis)
- The sensing electrodes (Z-axis) detect the Coriolis force that is produced by the mixture of driving momentum and external rotation.



Gyroscope

WSEN-ISDS: IMU SENSOR TECHNICAL PARAMETERS



Specifications

- ❑ Dimension - 2.5 mm x 3.0 mm x 0.86 mm (14pin) LGA package
- ❑ Acceleration Range: ±2g,/±4g,/±8g,/±16g
- ❑ Gyroscope Range: ±250 dps, ±500 dps, ± 1000 dps, ±2000 dps
- ❑ Resolution – 16 bits
- ❑ Sensitivity – 3%
- ❑ Current Consumption - HP mode: 690μA, NP mode: 370 μA, LP mode: 280 μA
- ❑ Low Noise Density – 75 μg/√Hz
- ❑ Output Data Rate - upto 6.6KHz, BW_{acc} – 1400 Hz, BW_{Gyro} – 937 Hz
- ❑ Operating Temperature Range: -40°C to +85°C

Application

- ❑ Industrial 4.0 and IoT connected devices
- ❑ Industry tools and factory equipment
- ❑ Antenna and platform stabilization
- ❑ Localization and navigation
- ❑ Robotics, drones and automation

Special Features

- ❑ Operation Mode – High Performance, Normal, Low Power
- ❑ 4 kbyte FIFO – flexible data handling
- ❑ 2 Independent Interrupts pins
- ❑ I^2C and SPI bus interface
- ❑ Embedded functions: freefall, wake-up, tap, activity, motion, orientation: 6D/4D, Tilt sensing